National University of Computer and Emerging

Artificial Intelligence Lab (AI 2002)

Date: March 17th, 2025

Course Instructor(s)
Ms. Ramsha Jat

Lab Mid Exam Paper A

Total Time: 90 minutes

Total Marks: 50

Total Questions: 03

Semester: Spring 2025

Campus: Karachi

Department: CS

Instructions:

- The Exam paper consists of 3 questions on 2 pages.
- Submit ipynb files name under your nu id with question number. E.g: k22-xxxx_Q1.ipynb.
- Submit ion will be on Software. Submission IP is **172.16.33.88:5000.**
- Enter your NU Id in Username and **Fast1234** in password.
- Read the question carefully before answering.

Student Name Roll No Section Student Signature

LLO # 1: [18 Marks]

Q1. A firefighter agent must rescue a trapped person (P) in a burning building represented as a grid. The grid contains safe rooms (0), fire (F), walls (#), the agent's starting position (A), and the trapped person (P). The agent can only move through safe rooms (0) and must use DFS to find the shortest path to the person. Identify the type of agent and implement DFS would be implemented to solve this problem. For the example input map below, show the path taken to reach the treasure.

```
A 0 0 # #
# F 0 # P
0 0 0 F 0
0 # F # 0
0 0 0 0 0
```

LLO # 3: [18 Marks]

Q2. Implement an A search algorithm* to find the shortest path in an N × M grid, where each cell contains a numerical value representing its movement cost. Some cells are blocked (#), meaning they cannot be traversed. The algorithm should determine the least-cost path from a given starting position (S) to a target position (T) while considering the terrain costs. The movement is restricted to four directions: up, down, left, and right, with the cost of movement determined by the value of the destination cell. The heuristic function should use the Manhattan distance to estimate the remaining cost to the goal. The implementation should be able to handle larger grids efficiently and return both the optimal path and the total travel cost from the start position to the target position.

S	1	2	#	3
1	#	2	#	#
1	2	1	2	#
#	#	1	#	#
1	1	1	2	T

LLO # 4: [14 Marks]

Q3. Implement a Genetic Algorithm (GA) to find the maximum value of the function .
$f(x) = 2x + 1$
within the range [0, 31].
Each possible solution should be represented as a 5-bit binary string (since 25=322^5 = 3225=32, covering values from 0 to 31). The algorithm should begin with a random population of binary-encoded values and evaluate their fitness using $f(x) = x^2 + x$. It should then apply natural selection to retain the fittest individuals, followed by single-point crossover to generate new offspring. After crossover, each offspring should undergo single-bit mutation, where one randomly selected bit is flipped with a small probability. The process should continue for a fixed number of generations or until convergence, returning the best x value that maximizes f(x) along with the function's maximum value.

National University of Computer and Emerging Sciences

Artificial Intelligence (AL-2002)

Lab Mid Exam

Total Time (hr): 1.5
Total Points: 50
Total Questions: 03

Paper A

Date: March 18th, 2025

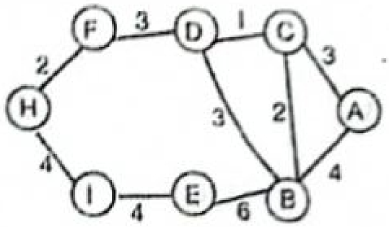
Course Instructor(s)

Mr. Talha Shahid. Miss. Alishba Subhani

Student Name | Roll No | Section | Student Signature

Question # 01 (LLO: 1) **[18 Points, 8 Weightage]**

In a densely connected city represented as a weighted graph, intersections are nodes and roads have varying lengths as edges. An agent perceives its environment by detecting nearby intersections and roads, each with a specific cost (distance). The agent's goal is to find the least-cost path between two locations using **Uniform Cost Search (UCS)**.



Agent's Starting Point: A | Agent's Goal: Reach I

Tasks:

1. Define the architecture of a goal-based agent, and environment class. **[07]**
2. Implement the **Uniform Cost Search (UCS)** function within the agent class to find the least-cost path. **[08]**
3. Create a `run_agent()` function that simulates the agent in the specified environment and prints the path and total cost. **[03]**

Question # 02 (LLO: 3) **[18 Points, 9 Weightage]**

In a **space exploration game**, a rover needs to navigate an alien planet's surface to find an energy module hidden at a specific location. The terrain is represented as a grid where:

- . represents a traversable path.
- # represents an obstacle.
- S represents the rover's starting point
- E represents the energy module's location.

The rover can move up, down, left, or right. Use *A Search** with the **Manhattan distance** to the energy module as a heuristic and the actual movement cost between cells.

S	.	#	.	E
.	#	.	#	.
.	.	.	#	.
#	.	#	.	#
.	.	.	.	.

S = Rover Starting Point
E = Energy module's location
. = Traversable Path
# = Obstacle

Task:

1. Define the heuristic function which calculates the Manhattan distance between a point and the energy module. ($|x1 - x2| + |y1 - y2|$). **[03]**
2. Define a `generate_neighbours` function which provides all possible neighbours of a given state based on the possible moves of the rover. **[04]**
3. Implement the ***A* function*** which finds the shortest path from the start to the energy module, considering both the heuristic cost and the actual cost. **[08]**
4. A main function which calls the A* Search function and displays the shortest path taken by the rover, the total number of steps, and the total cost. **[03]**

Question # 03 (LLO: 4) **[14 Points, 7 Weightage]**

Genetic Algorithm for the 0/1 Knapsack Problem |refer to the provided guide.

A logistics company is optimizing its delivery truck space by selecting the most valuable set of packages that fit within a weight limit. Each package has a specific weight and value, and the company needs to maximize total value while ensuring the total weight does not exceed the truck's capacity.

To solve this problem, you will implement a **Genetic Algorithm (GA)**. Some parts of the algorithm are already provided, and your task is to complete the missing functions.

You **must implement** the following functions to complete the GA:

1. **`create_random_individual(num_items)`**
   - Generate a random binary list representing a possible solution (1 = item included, 0 = item excluded).
2. **`initialize_population(pop_size, num_items)`**
   - Create an initial population of pop_size individuals using create_random_individual().
3. **`select_parents(population, fitness_scores)`**
   - Select the top 50% of the population based on fitness for reproduction.
4. **`crossover(parent1, parent2)`**
   - Implement **single point** crossover.

National University of Computer and Emerging Sciences

Artificial Intelligence Lab (AI1002)

Lab Mid Exam A

Total Time: 90 minutes
Total Points: 50
Total Questions: 03

Date: March 19th 2025

Semester: Spring 2025
Campus: Karachi
Department: CS, AI, SE, CY &DS

Course Instructor(s)
Ms Sarah , Mr Shafique, Mr. Muhammad Khalid, Mr Abdullah Yaqoob

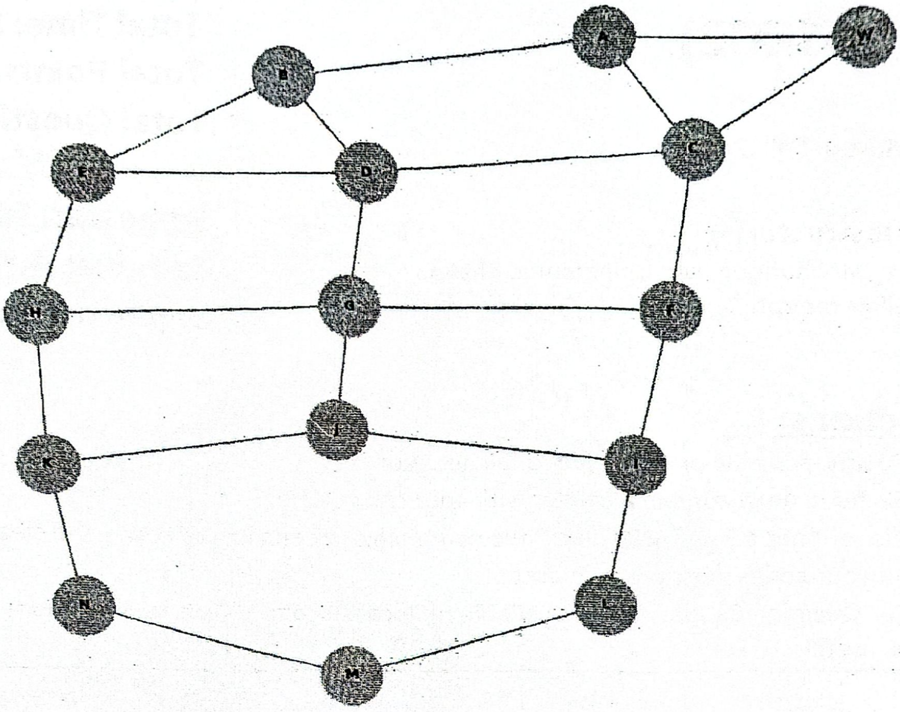
Instructions:

- Submit a .py file or .ipynb file for all questions.
- Each file must contain a header with your roll number.
- Also submit a ".pdf" OR ".docx" file containing screenshots of codes and clearly show the output.
- All your codes must produce output.
- For Question 03, the code is available on local storage (\\172.16.5.43\Exam Material\ (Your Teacher Name)).

Student Name ______ Roll No ______ Section ______ Student Signature ______

LLO # 1: Explain and implement BFS and DFS algorithms, comparing their efficiency and applications in uninformed search. [18 points]

Q1: You are part of a logistics company that employs goal-based agents for efficient route planning in a city. The city map is represented as an undirected graph, where each node represents a delivery location, and the edges represent possible routes between those locations. The company's goal is to design a **goal-based agent** that finds the path between a starting location; warehouse (W) and the destination (N). You need to use **Breadth-First Search (BFS)** to solve this.



LLO #3: Apply and implement A* and GBFS algorithms, evaluating their heuristics, efficiency, and applications in informed search. ***[18 points]***

Q2. A robotic agent is deployed inside a structured environment represented as a **6x6 grid-based maze**. The robot's goal is to navigate from a given **starting position (0,0)** to a designated **exit point (4,3)** while minimizing the total distance traveled using Manhattan distance.
However, the maze contains **obstacles** (represented as `1`s in the grid), meaning the robot cannot pass through those cells. The robot can move **up, down, left, or right** but cannot move diagonally. The pathfinding must be efficient, ensuring the shortest possible route is taken using the *Greedy Best first search*

The grid look like

```
grid = [
  [0, 0, 0, 0, 0, 0],
  [0, 1, 1, 1, 0, 0],
  [0, 0, 0, 1, 0, 0],
  [0, 1, 0, 1, 0, 0],
  [0, 1, 0, 0, 0, 0],
]
```

*LLO #4: **Implement Genetic Algorithms to solve optimization problems, analyzing their operators, convergence, and performance.*** ***[14 points]***

Q3.

Instructions:

The code is available on local storage (\\172.16.5.43\Exam Material\ (Your Teacher Name)).

Problem Statement:

In a computer network, several data packets need to be routed through a series of routers to reach their destination. However, due to network congestion, some routers may experience higher loads than others, resulting in delays and inefficiencies. Your task is to implement a **Genetic Algorithm (GA)** to optimize the routing of data packets, minimizing the total network congestion and ensuring efficient data delivery.

You are given a network of routers and a set of possible routes between them. Each route has a bandwidth and a congestion level associated with it. The objective is to find the optimal routing paths for a set of packets such that the total network load (measured by the congestion levels of the routes) is minimized.

The tasks are to be completed with pop_size = 10 and for 50 generations.

Task:

1. By using Roulette Selection method (individuals are selected for reproduction based on their fitness, with higher fitness individuals having a higher probability of being chosen), create a selection function. [4]
2. You are expected to write a crossover function using Single Point crossover. [4]
3. You are expected to write a mutate function using Binary Mutation (a randomly selected bit in an individual's binary representation is flipped to introduce variation). [4]
4. Update such that the code runs completely and outputs best solution. [2]